

Bayesian Classification

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Outline

- ▶ Introduction
- ▶ Bayes Theorem
- ▶ Finding MAP hypothesis
- ▶ Naive Bayes Classifier
- ▶ Bayesian belief networks

Introduction

- ▶ Goal:
 - ▶ Determine the most probable hypothesis (class)
 - ▶ E.g, Given new instance x , what is its most probable classification?
- ▶ Probabilistic learning and prediction:
 - ▶ Estimate explicit probabilities for all hypotheses (classes)
 - ▶ Predict multiple hypotheses, weighted by their probabilities
 - ▶ Can combine prior knowledge (such as prior probabilities, probability distributions, causal relationships between variables in belief networks) with observed data

Introduction (*Cont'd*)

- ▶ Incremental learning:
 - ▶ Each training example can incrementally increase/decrease the probability that a hypothesis is correct.
- ▶ Flexible in handling inconsistency
- ▶ Standard:
 - ▶ provides a standard of optimal decision making against which other methods can be measured

Bayes Theorem

$$P(h | x) = \frac{P(x | h)P(h)}{P(x)}$$

- ▶ $P(h)$ = prior probability of hypothesis h
- ▶ $P(x)$ = probability that example x is observed
- ▶ $P(h | x)$ = posterior probability of h given x
- ▶ $P(x | h)$ = conditional probability of x given h (often called the likelihood of h given x)

Finding Maximum a Posteriori Hypothesis

$$P(h | x) = \frac{P(x | h)P(h)}{P(x)}$$

- ▶ Goal: Find the most probable hypothesis h from a set H of candidate hypotheses given an example x .
- ▶ The most probable hypothesis is called *maximum a posteriori (MAP)* hypothesis h_{MAP} :

$$h_{MAP}(x) = \arg \max_{h \in H} P(h | x)$$

$$= \arg \max_{h \in H} \frac{P(x | h)P(h)}{P(x)} \quad (P(x) \text{ is constant for all hypotheses})$$

$$= \arg \max_{h \in H} P(x | h)P(h)$$

- ▶ If assume $P(h_i) = P(h_j)$ (classes are equally likely), then can further simplify, and choose the *Maximum likelihood* (ML) hypothesis

$$h_{ML}(x) = \arg \max_{h \in H} P(x | h)$$

Example

- ▶ Does patient have cancer or not?
 - ▶ A patient takes a lab test and the result comes back positive.
 - ▶ The test returns a correct positive result in only 98% of the cases in which the disease is actually present,
 - ▶ The test returns a correct negative result in only 97% of the cases in which the disease is not present.
 - ▶ Furthermore, .008 of the entire population have this cancer.

$$P(cancer) =$$

$$P(\neg cancer) =$$

$$P(+ | cancer) =$$

$$P(- | cancer) =$$

$$P(+ | \neg cancer) =$$

$$P(- | \neg cancer) =$$

Our goal is to find the maximum between:

$$P(cancer | +) \text{ and } P(\neg cancer | +)$$

Learning Probabilities from Data

- Suppose we do not know the probabilities used in the example in the last slide.
- But we are given a set of data.
- In order to conduct the reasoning, i.e., to find the MAP hypothesis h_{MAP} , we can estimate the probabilities used in the reasoning from the data.

Learning Probabilities from Data

- Suppose there are k possible hypotheses (i.e., classes):
$$h_1, h_2, \dots, h_k$$

- In order to find:

$$h_{MAP}(x) = \arg \max_{h_i \in H} P(x | h_i) P(h_i)$$

- We need to estimate:
 - ▶ $P(h_1), P(h_2), \dots, P(h_k),$
 - ▶ $P(x|h_1), P(x|h_2), \dots, P(x|h_k)$ for each possible instance x ,

Practical Problem with Finding MAP Hypothesis

- ▶ Suppose instance x is described by attributes values $\langle x_1, x_2, \dots, x_n \rangle$ and there is a set C of classes: $c_1, c_2, \dots, c_k$.

$$\begin{aligned} c_{MAP}(x) &= \arg \max_{c_j \in C} P(c_j | x_1, x_2, \dots, x_n) \\ &= \arg \max_{c_j \in C} \frac{P(x_1, x_2, \dots, x_n | c_j) P(c_j)}{P(x_1, x_2, \dots, x_n)} \\ &= \arg \max_{c_j \in C} P(x_1, x_2, \dots, x_n | c_j) P(c_j) \end{aligned}$$

- ▶ Given data set with many attributes, it is infeasible to estimate $P(x_1, x_2, \dots, x_n | c_j)$ for all possible x values unless we have a very, very large set of training data. It is also computationally expensive.

Naive Bayes Classifier

- ▶ Naive assumption: values of attributes are conditionally independent given a class

$$P(x_1, x_2, \dots, x_n | c_j) = \prod_i P(x_i | c_j)$$

which gives:

$$\begin{aligned} c_{NB}(x) &= \arg \max_{c_j \in C} P(x_1, x_2, \dots, x_n | c_j) P(c_j) \\ &= \arg \max_{c_j \in C} P(c_j) \prod_i P(x_i | c_j) \end{aligned}$$

- ▶ Probabilities can be estimated from the training data.

Estimating Probabilities

- ▶ Estimate $P(c_j)$:

$$P(c_j) = \frac{\text{\# of training examples of class } c_j}{\text{\# of training examples}}$$

- ▶ Estimate $P(x_i|c_j)$ for each attribute value x_i of attribute A_i and each class c_j
 - ▶ If attribute A_i is categorical,

$$P(x_i | c_j) = \frac{\text{\# of training examples of class } c_j \text{ with } x_i \text{ for } A_i}{\text{\# of training examples of class } c_j}$$

Estimating Probabilities

- ▶ If attribute A_i is continuous, can assume a distribution, such as normal distribution,

$$P(x_i | c_j) = \frac{1}{\sqrt{2\pi}\sigma_{c_j}} e^{-\frac{(x_i - \mu_{c_j})^2}{2\sigma_{c_j}^2}}$$

where μ_{c_j} and σ_{c_j} are the mean and standard deviation of the values of A_i for training examples of class c_j

$$\sigma_{c_j} = \sqrt{\frac{1}{n-1} \sum_{x_i \in c_j} (x_i - \mu_{c_j})^2}$$

Naive Bayes Algorithm

- ▶ Naive Bayes Learning (*from examples*)

- ▶ For each class c_j

$$\hat{P}(c_j) \leftarrow \text{estimate } P(c_j)$$

- ▶ For each attribute for which x_i is a value

$$\hat{P}(x_i | c_j) \leftarrow \text{estimate } P(x_i | c_j)$$

- ▶ Classifying new instance (x)

$$c_{NB}(x) = \arg \max_{c_j \in C} \hat{P}(c_j) \prod_{x_i \in x} \hat{P}(x_i | c_j)$$

Example

Training dataset

Classes:

c1:buys_computer='yes'

c2:buys_computer='no'

Classify new example:

X = (age ≤ 30,
Income = medium,
Student = yes
Credit_rating = Fair)

age	income	student	credit_rating	buys_computer
≤30	high	no	fair	no
≤30	high	no	excellent	no
30...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤30	medium	no	fair	no
≤30	low	yes	fair	yes
>40	medium	yes	fair	yes
≤30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Example (*cont'd*)

► Learning:

► Compute $P(c_i)$

$$P(\text{buy_computer}=\text{"yes"})=9/14$$

$$P(\text{buy_computer}=\text{"no"})=5/14$$

► Compute $P(x_j|c_i)$ for each class and each attribute value pair:

$$P(\text{age} \leq 30 \mid \text{buys_computer}=\text{"yes"}) = 2/9 = 0.222$$

$$P(\text{age} \leq 30 \mid \text{buys_computer}=\text{"no"}) = 3/5 = 0.6$$

.....

$$P(\text{income}=\text{"medium"} \mid \text{buys_computer}=\text{"yes"}) = 4/9 = 0.444$$

$$P(\text{income}=\text{"medium"} \mid \text{buys_computer}=\text{"no"}) = 2/5 = 0.4$$

.....

$$P(\text{student}=\text{"yes"} \mid \text{buys_computer}=\text{"yes"}) = 6/9 = 0.667$$

$$P(\text{student}=\text{"yes"} \mid \text{buys_computer}=\text{"no"}) = 1/5 = 0.2$$

.....

$$P(\text{credit_rating}=\text{"fair"} \mid \text{buys_computer}=\text{"yes"}) = 6/9 = 0.667$$

$$P(\text{credit_rating}=\text{"fair"} \mid \text{buys_computer}=\text{"no"}) = 2/5 = 0.4$$

.....

Example (*cont'd*)

- Classification: to classify:

$x = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$

$P(x|c_i)$:

$$\begin{aligned} & P(x|\text{buys_computer}=\text{"yes"}) \\ &= P(\text{age} \leq 30 | \text{buys_computer}=\text{yes}) \times P(\text{income}=\text{medium} | \text{buys_computer}=\text{yes}) \times \\ & \quad P(\text{student}=\text{yes} | \text{buys_computer}=\text{yes}) \times P(\text{credit}=\text{fair} | \text{buys_computer}=\text{yes}) \\ &= 0.222 \times 0.444 \times 0.667 \times 0.0667 \\ &= 0.044 \end{aligned}$$

$$P(x|\text{buys_computer}=\text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$$

$P(c_i|x) \propto P(x|c_i) * P(c_i)$:

$$P(\text{buys_computer}=\text{"yes"}|x) \propto$$

$$P(x|\text{buys_computer}=\text{"yes"}) * P(\text{buys_computer}=\text{"yes"}) = 0.028$$

$$P(\text{buys_computer}=\text{"no"}|x) \propto$$

$$P(x|\text{buys_computer}=\text{"no"}) * P(\text{buys_computer}=\text{"no"}) = 0.007$$

x belongs to class "buys_computer=yes"

Naïve Bayesian Classifier: Comments

- ▶ Advantages :

- ▶ Easy to implement
- ▶ Good results were obtained in many applications

- ▶ Disadvantage

- ▶ Assumption: class conditional independence of attributes, therefore loss of accuracy
- ▶ Practically, dependencies exist among attributes
 - ▶ For example, *headache* and *body temperature* are dependent attributes for *flu* dataset.
- ▶ Dependencies among these cannot be modeled by Naïve Bayesian Classifier

- ▶ How to deal with these dependencies?

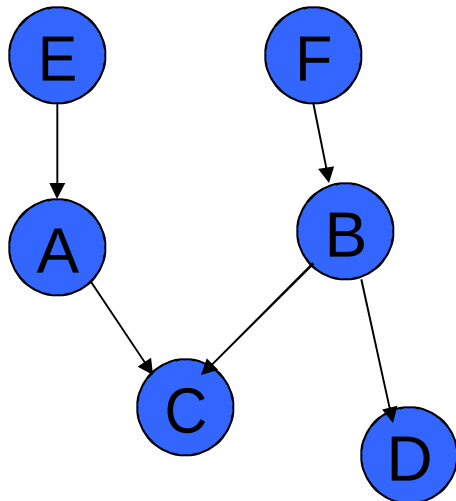
- ▶ Bayesian Belief Networks

Bayesian Belief Networks

- ▶ Naive Bayes assumption of conditional independence is too restrictive.
- ▶ But it's intractable without such assumptions...
- ▶ Bayesian Belief networks provide an intermediate approach which
 - ▶ allows dependencies among attributes
 - ▶ but assumes conditional independence among subsets of attributes.

Bayesian Belief Networks

- ▶ A graphical model of causal relationships. Two components:
 - ▶ A *directed acyclic graph (DAG)*: represents dependency among variables (attributes)

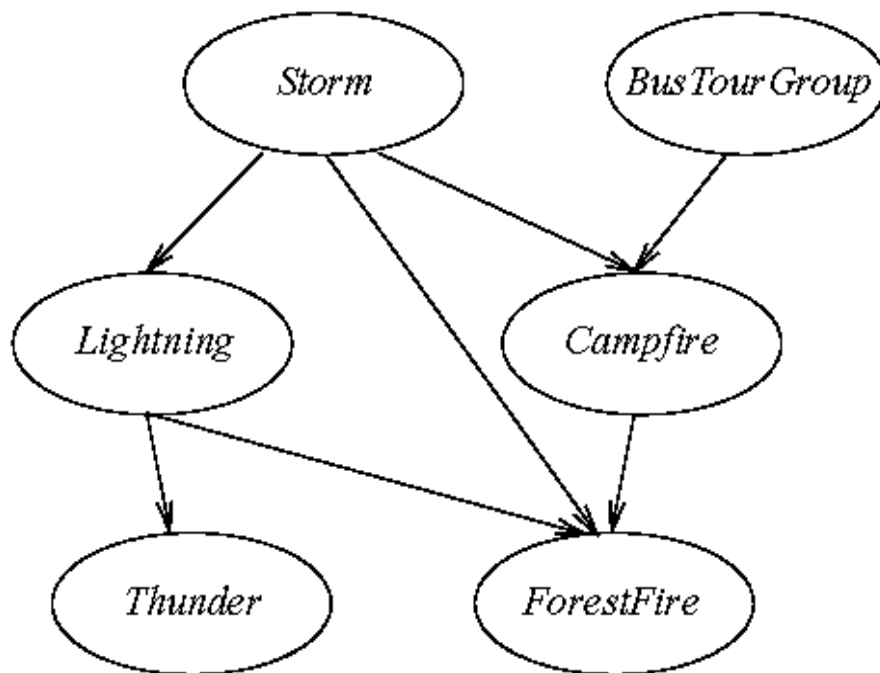


- **Nodes:** variables (including class attribute)
- **Links:** dependencies (e.g., A depends on E)
- **Parents:** immediate predecessors. E.g., A,B are the parents of C. B is the parent of D
- **Descendant:** X is a descendant of Y if there is a direct path from Y to X.
- **Conditional Independency:**
 - Assume: each variable is conditionally independent of its nondescendants given its parents.
 - Definition: X is conditionally independent of Y given Z iff $P(X|Y,Z)=P(X|Z)$
 - E.g.: C is conditional independent of D given A and B. Thus, $P(C|A, B, D)=P(C|A, B)$
- **Acyclic:** has no loops or cycles

- ▶ A *conditional probability table (CPT)* for each variable X: specifies the conditional probability distribution $P(X|\text{Parents}(X))$.

Example of CPT

- Suppose each variable is binary (contain two values: X and $\neg X$)



CPT table for *Campfire*

	S, B	$S, \neg B$	$\neg S, B$	$\neg S, \neg B$
C	0.4	0.1	0.8	0.2
$\neg C$	0.6	0.9	0.2	0.8

$P(C|S, B)$

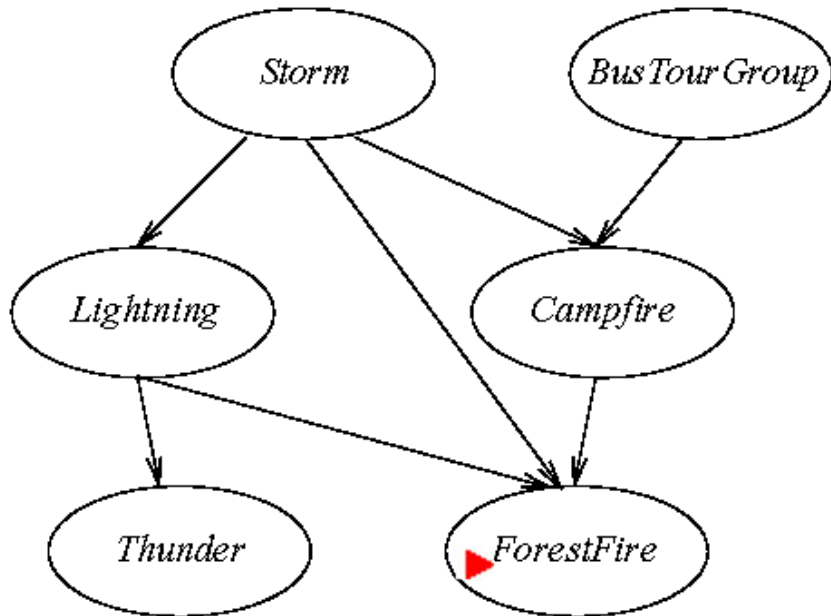


- There is a conditional probability table (CPT) for each variable

Inference Rule in Bayesian Networks

- ▶ The joint probability of any tuple $(x_1, \dots, x_n)$ corresponding to the variables or attributes $(X_1, \dots, X_n)$ is computed by

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid \text{Parents}(X_i))$$

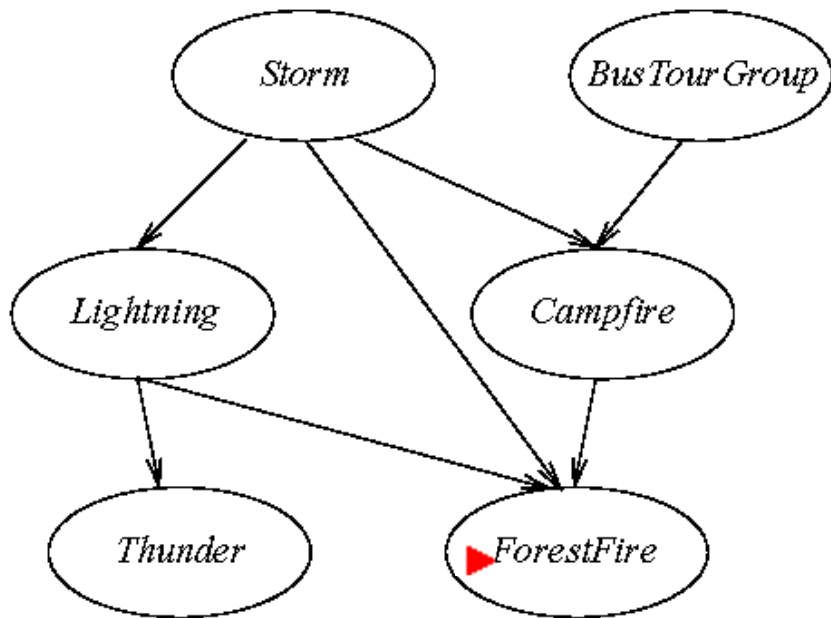


- ▶ Example:

$$P(\neg S, B, \neg L, C, \neg T, F) = P(\neg S) \times P(B) \times P(\neg L \mid \neg S) \times P(C \mid \neg S, B) \times P(\neg T \mid \neg L) \times P(F \mid \neg L, \neg S, C)$$

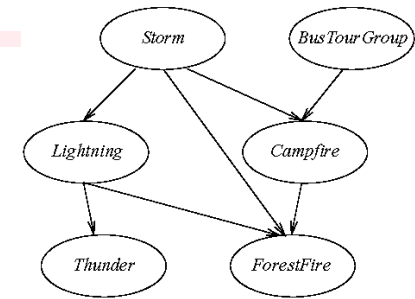
Inference in Bayesian Networks

- ▶ A Bayesian network can be used to infer the (probabilities of) values of one or more network variables, given observed values of others.
- ▶ Example:
 - ▶ Given Storm=0, BusTourGroup=1, Lightning=0, Campfire=1, Thunder=0, we want to know ForestFire=?



- ▶ Compute two probabilities:
 - (1) $P(F | \neg S, B, \neg L, C, \neg T) = P(F | \neg L, \neg S, C)$
 - (2) $P(\neg F | \neg S, B, \neg L, C, \neg T) = P(\neg F | \neg L, \neg S, C)$
- ▶ ForestFire = True if (1) > (2)

Inference in Bayesian Networks



▶ Another example:

- ▶ Given Storm=1, Campfire=0, ForestFire=1, what is the probability distribution of Thunder?
- ▶ Compute two probabilities:

$$(1) P(T | S, \neg C, F) = P(T, L | S, \neg C, F) + P(T, \neg L | S, \neg C, F)$$

$$= P(T | L, S, \neg C, F)P(L | S, \neg C, F) + P(T | \neg L, S, \neg C, F)P(\neg L | S, \neg C, F)$$

$$= P(T | L)P(L | S, \neg C, F) + P(T | \neg L)P(\neg L | S, \neg C, F)$$

$$\text{where } P(L | S, \neg C, F) = \frac{P(L, F | S, \neg C)}{P(F | S, \neg C)} = \frac{P(F | L, S, \neg C)P(L | S, \neg C)}{P(F, L | S, \neg C) + P(F, \neg L | S, \neg C)}$$

$$= \frac{P(F | L, S, \neg C)P(L | S)}{P(F, L | S, \neg C) + P(F, \neg L | S, \neg C)} = \frac{P(F | L, S, \neg C)P(L | S)}{P(F | L, S, \neg C)P(L | S, \neg C) + P(F | \neg L, S, \neg C)P(\neg L | S, \neg C)}$$

$$= \frac{P(F | L, S, \neg C)P(L | S)}{P(F | L, S, \neg C)P(L | S) + P(F | \neg L, S, \neg C)P(\neg L | S)}$$

$$\text{and similarly } P(\neg L | S, \neg C, F) = \frac{P(F | \neg L, S, \neg C)P(\neg L | S)}{P(F | L, S, \neg C)P(L | S) + P(F | \neg L, S, \neg C)P(\neg L | S)}$$

(2) $P(\neg T | S, \neg C, F)$ can be calculated similarly.

- ▶ Thunder = True if (1) > (2)

Learning of Bayesian Networks

- ▶ Several scenarios of this learning task
 - ▶ Network structure might be *known* or *unknown*.
 - ▶ Training examples might provide values of all network variables, or just *some*.
- ▶ Scenario 1: If structure known and observe all variables:
 - ▶ Then it's easy as training a Naive Bayes classifier.
 - ▶ Learn only CPTs (estimate the conditional probabilities from training data)

Learning of Bayesian Networks

- ▶ Scenario 2: Suppose structure known, variables partially observable
 - ▶ For example, observe *ForestFire*, *Storm*, *BusTourGroup*, *Thunder*, but not *Lightning*, *Campfire*...
 - ▶ Similar to training neural network with hidden units. In fact, can learn network conditional probability tables using *gradient ascent* method!
- ▶ Scenario 3: When structure unknown
 - ▶ Use heuristic search or constraint-based technique to search through potential structures.
 - ▶ K2 algorithm

Summary: Bayesian Belief Networks

- ▶ Combine prior knowledge with observed data
- ▶ Intermediate approach that allows both dependencies and conditional independencies
- ▶ Other issues
 - ▶ Extend from categorical to real-valued variables
 - ▶ Parameterized distributions instead of tables
 - ▶ More effective inference and learning methods
 - ▶ ...